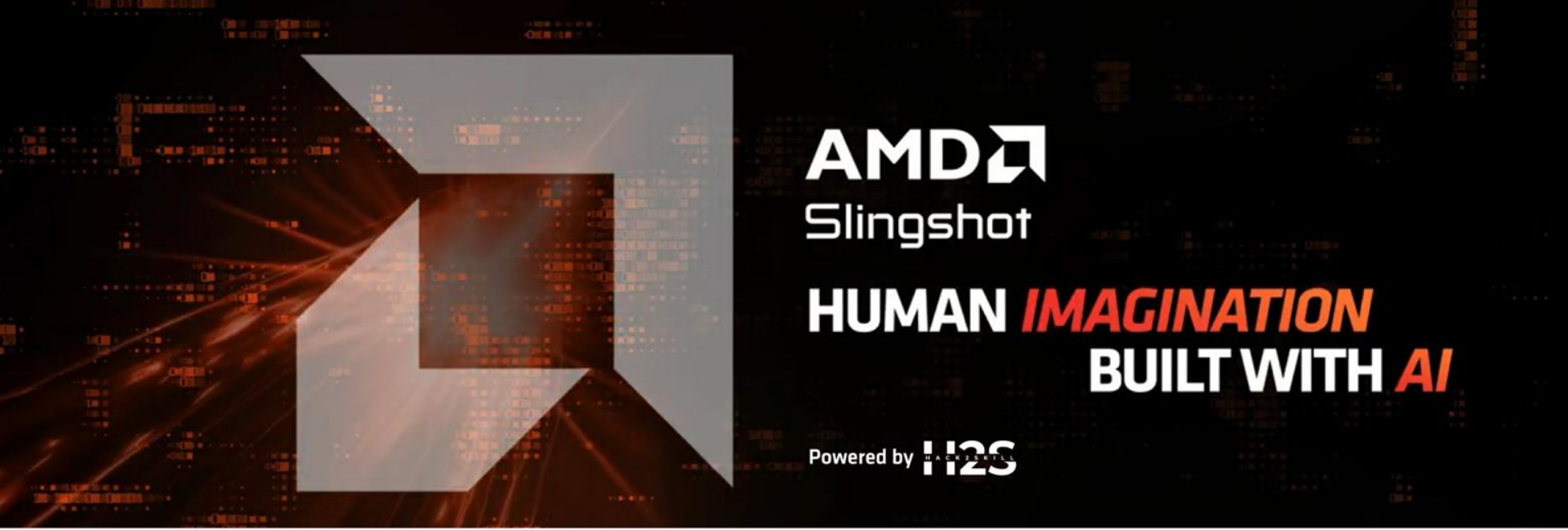


Team Details

- ➔ Team name – Intellix
- ➔ Team leader name –S.Megha Sree
- ➔ Problem Statement - Campuses lack real-time, actionable intelligence to detect and prevent energy and water wastage

Brief about the idea

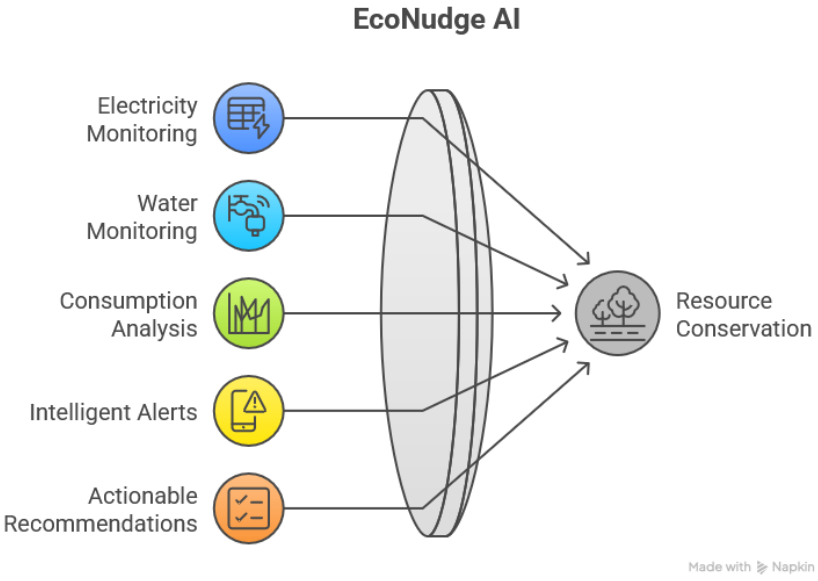
EcoNudge AI - Smart Campus Energy & Water Intelligence Platform

EcoNudge AI is a **smart sustainability dashboard** designed to help educational campuses **monitor, analyze, and reduce electricity and water wastage in real time**.

EcoNudge AI -

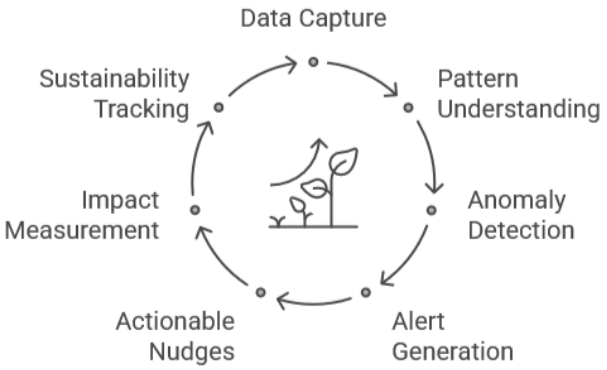
→ ”Doesn’t just monitor resources
— but nudge people to act at the right time.”

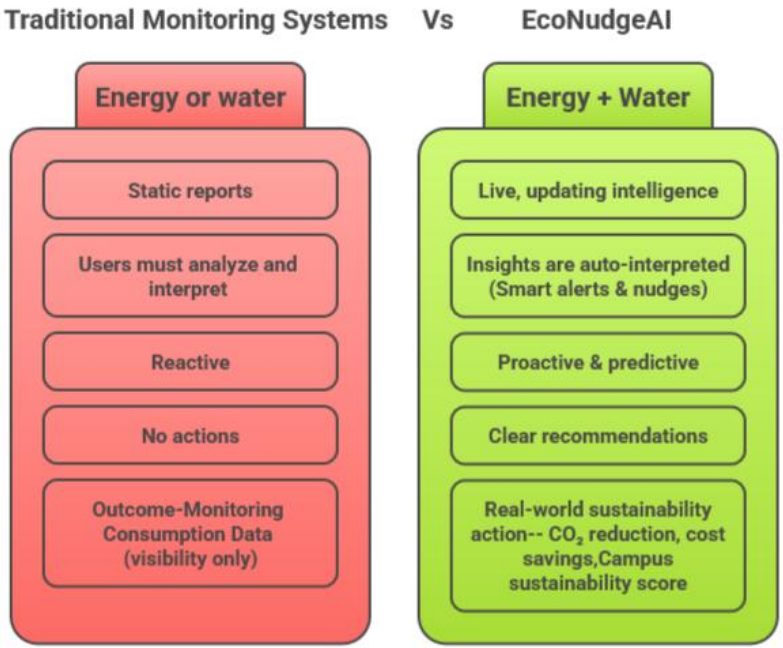
→bridges the gap between **data visibility and real-world action**, enabling campuses to move towards **measurable, practical sustainability**.

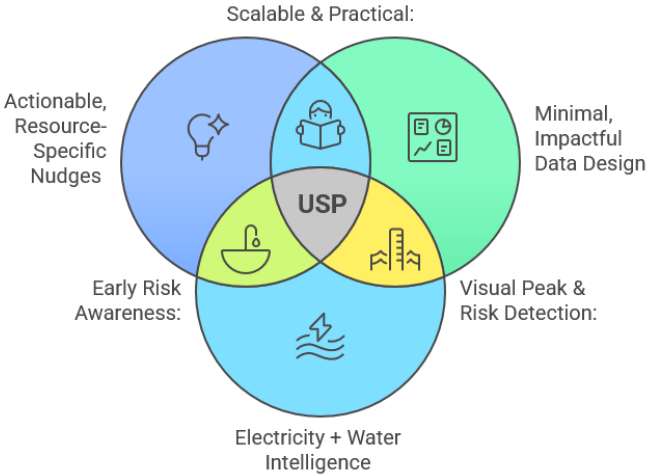


Opportunities

EcoNudge AI shifts the mindset from “monitoring” to “decision-making”.







List of features offered by the solution

Solution Features

Smart Monitoring

(Based on time,Location) Tracks electricity and water usage, location-wise, with continuous visualization.

AI Insights & Actionable Nudges

Classifies abnormal usage and provides context-aware suggestions for reduction.

Sustainability Impact Estimation

Estimates cost savings and CO_2 reduction to quantify sustainability benefits.

Future-Ready Design

Ready for IoT sensor integration, predictive analytics, and campus-wide expansion.

Intelligent Alert System

Detects high usage, overuse, and leaks, prioritizing alerts by severity.

Peak Usage Analysis

Identifies peak electricity and water consumption to aid in planning.

Clean & Intuitive Dashboard

Minimal clutter, color-coded alerts, and smooth visual hierarchy for easy interpretation.

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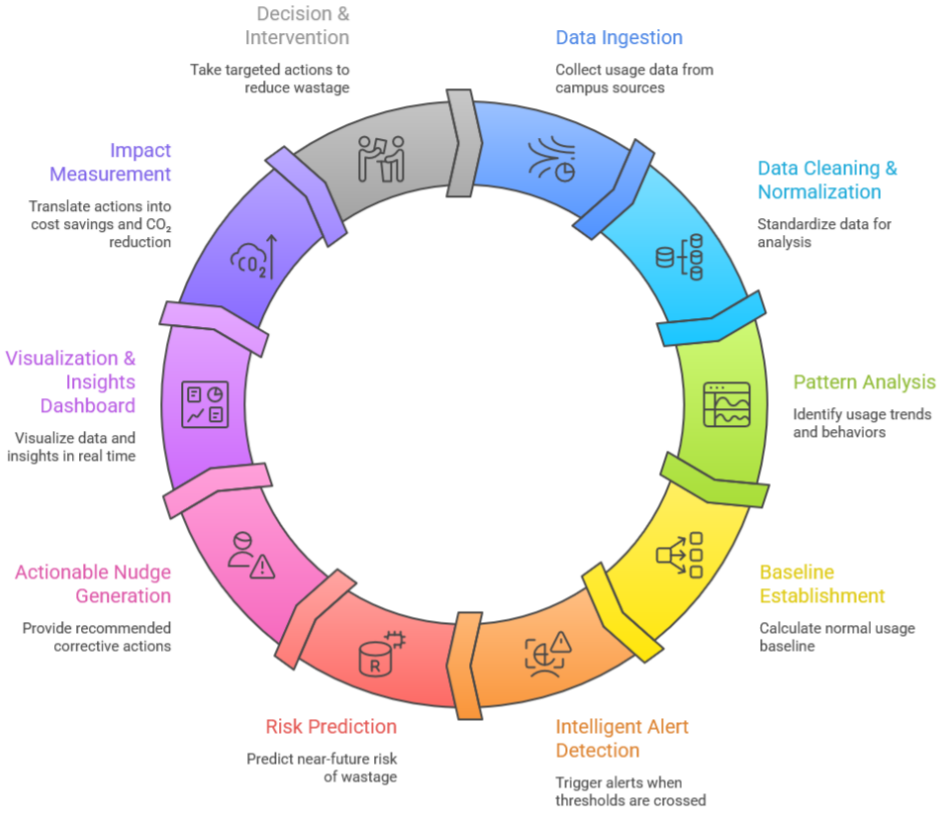
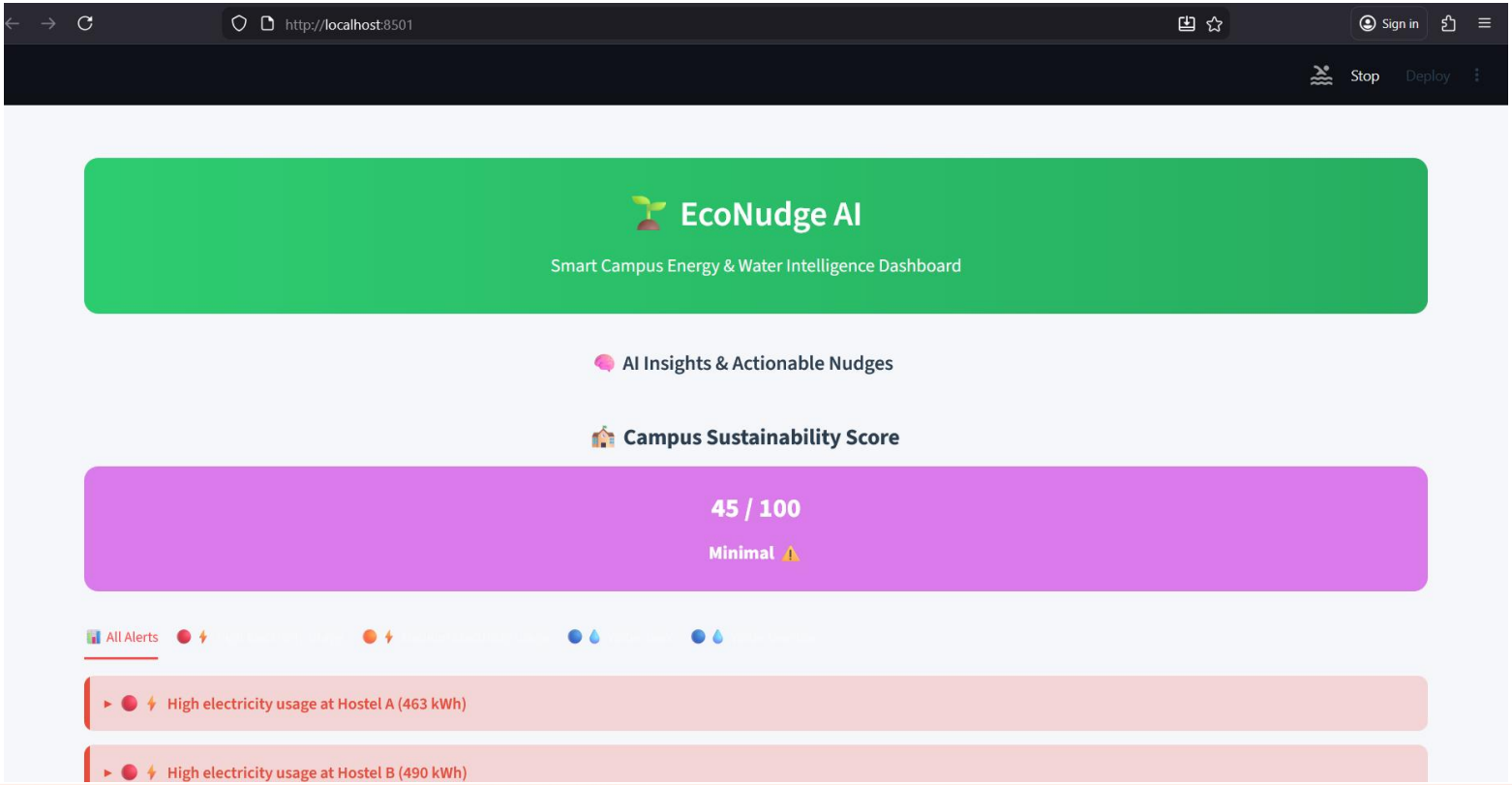
Process flow diagram or Use-case diagram

EcoNudge AI Process Flow

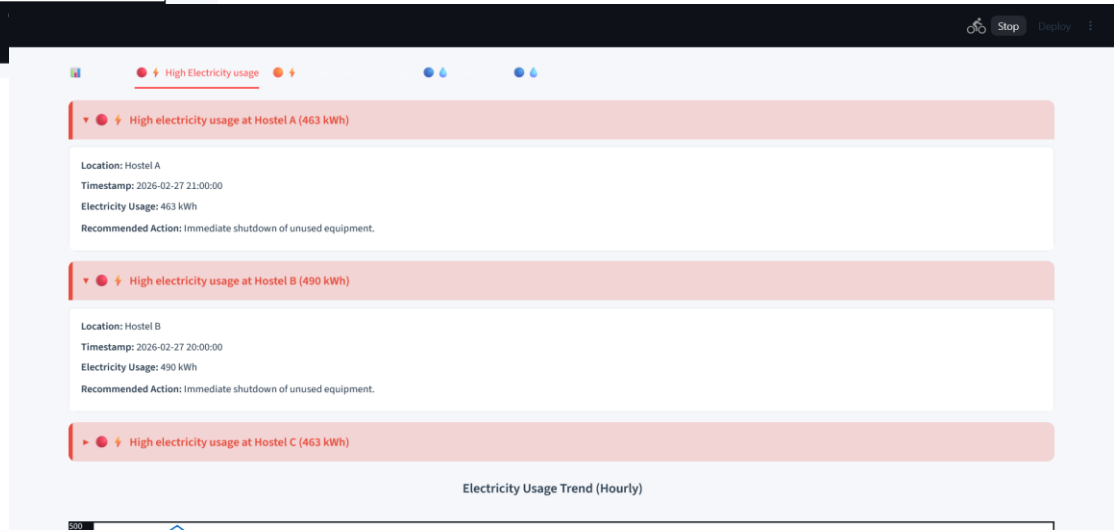


Usecase diagram

Wireframes/Mock diagrams of the proposed solution - Screenshot of dashboard created



Stop
Deploy
High Electricity usage
High electricity usage at Hostel A (463 kWh)
High electricity usage at Hostel B (490 kWh)
High electricity usage at Hostel C (463 kWh)
Electricity Usage Trend (Hourly)
500
400
300
200
100
0
06 PM 07 PM 08 PM 09 PM 10 PM 11 PM Sat 28 01 AM 02 AM 03 AM 04 AM 05 AM 06 AM 07 AM 08 AM 09 AM 10 AM 11 AM 12 PM
Hostel A
Hostel B
Hostel C
Lab 1


Stop
Deploy
High Electricity usage
High electricity usage at Hostel A (463 kWh)
Location: Hostel A
Timestamp: 2026-02-27 21:00:00
Electricity Usage: 463 kWh
Recommended Action: Immediate shutdown of unused equipment.
High electricity usage at Hostel B (490 kWh)
Location: Hostel B
Timestamp: 2026-02-27 20:00:00
Electricity Usage: 490 kWh
Recommended Action: Immediate shutdown of unused equipment.
High electricity usage at Hostel C (463 kWh)
Electricity Usage Trend (Hourly)
500

Deploy

All Alerts

- High electricity usage at Hostel A (463 kWh)
- High electricity usage at Hostel B (490 kWh)
- High electricity usage at Hostel C (463 kWh)
- Possible water leakage at Hostel C (9200 L)
- Elevated electricity usage at Lab 1 (312 kWh)
- Elevated electricity usage at Hostel A (316 kWh)
- Elevated electricity usage at Hostel B (328 kWh)
- Elevated electricity usage at Hostel C (328 kWh)

Electricity Usage Trend (Hourly)

Stop Deploy

EcoNudge AI

Smart Campus Energy & Water Intelligence Dashboard

AI Insights & Actionable Nudges

Campus Sustainability Score

45 / 100

Minimal ⚠️

Water Leak

Possible water leakage at Hostel C (9200 L)

Location: Hostel C

Timestamp: 2026-02-28 14:00:00

Water Usage: 9200 liters

Recommended Action: Immediate pipeline inspection.

AMD

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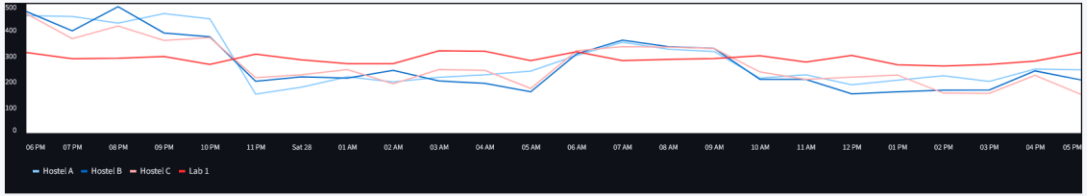
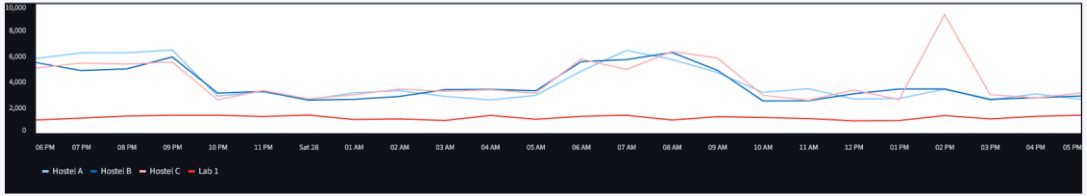
Stop Deploy

Electricity Usage Trend (Hourly)

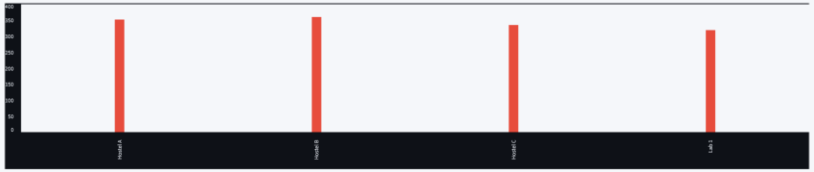


Water Usage Trend (Hourly)

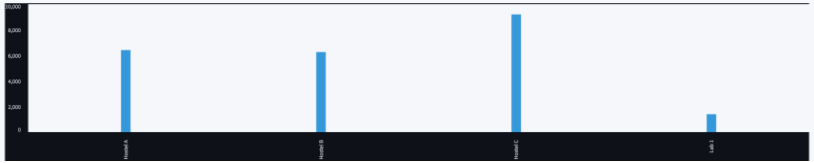


Stop Deploy

Peak Electricity Consumption by Location

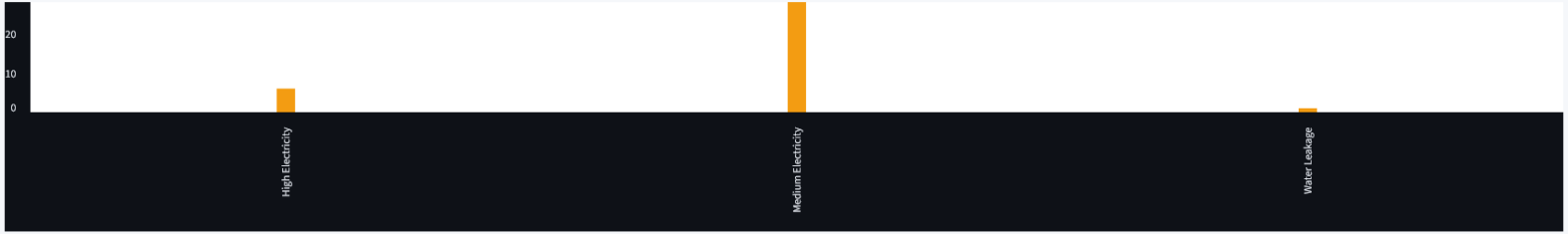


Peak Water Consumption by Location



Stop Deploy

Alert Distribution (AI Insights)

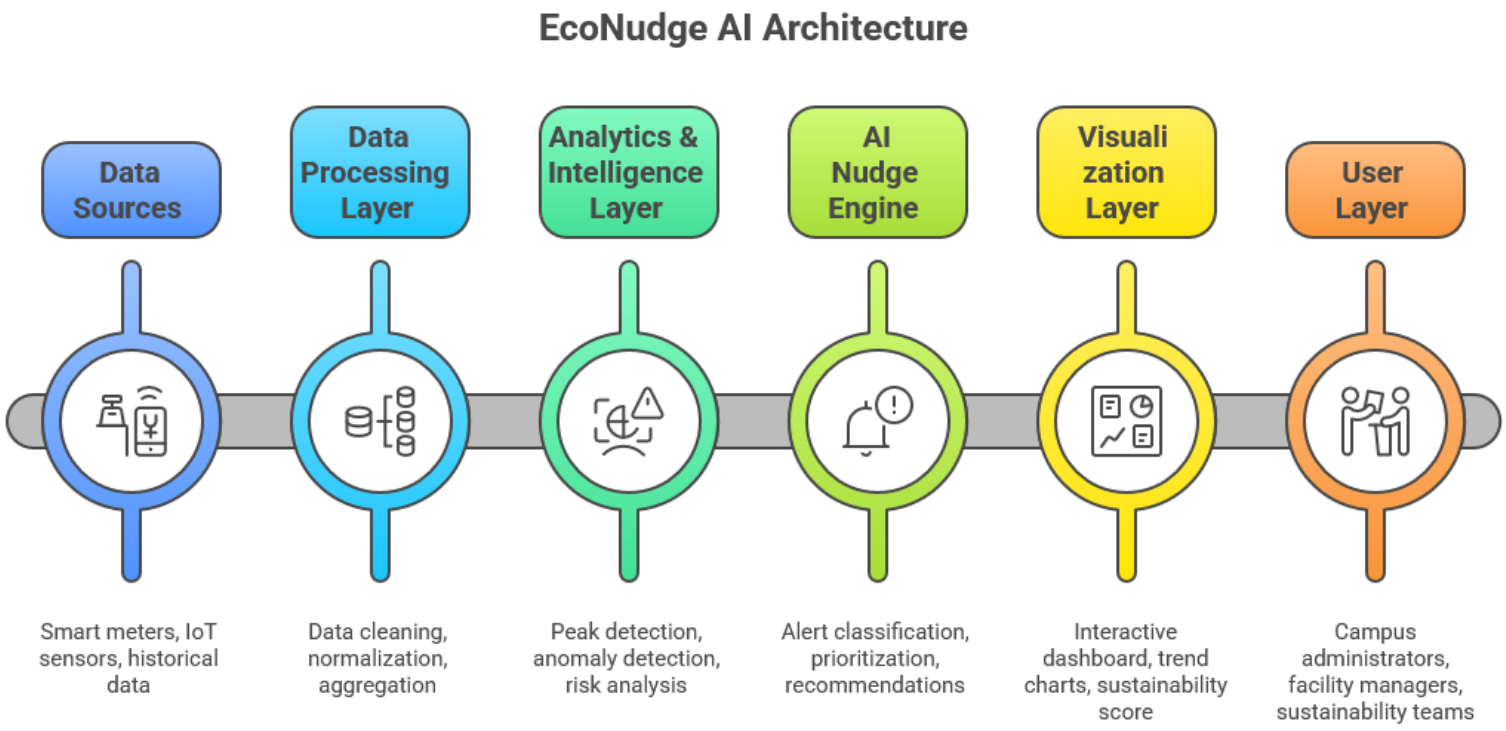


Sustainability Impact & Savings Estimator

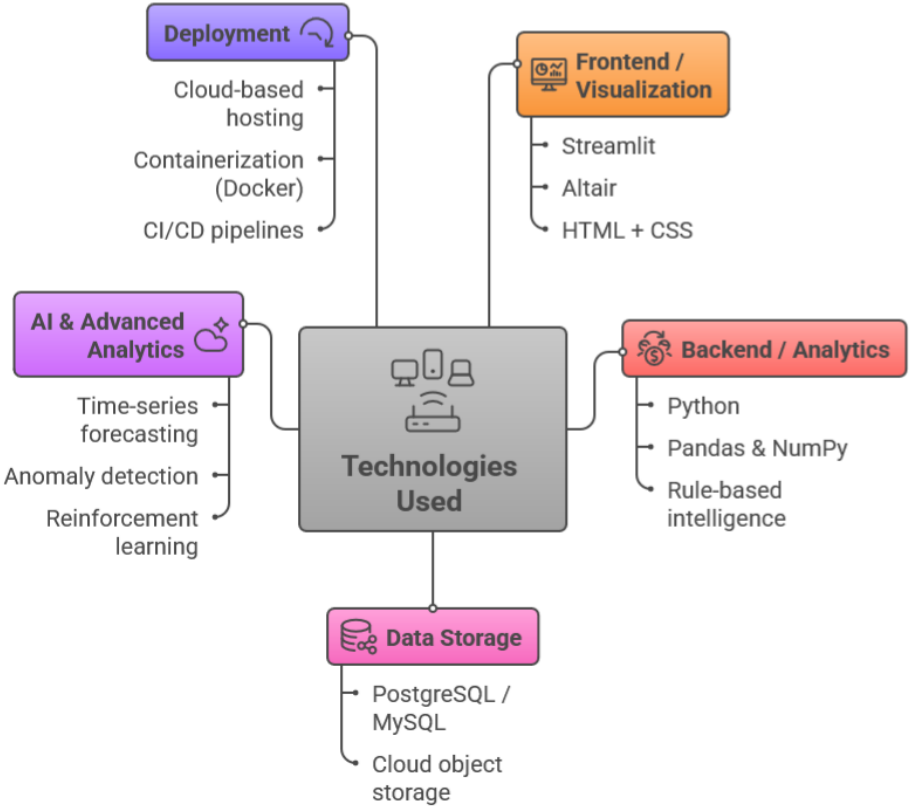
10

💰 Estimated Annual Cost Savings: ₹12000
🌱 Estimated CO_2 Reduction: 550 kg/year

Architecture diagram of the proposed solution

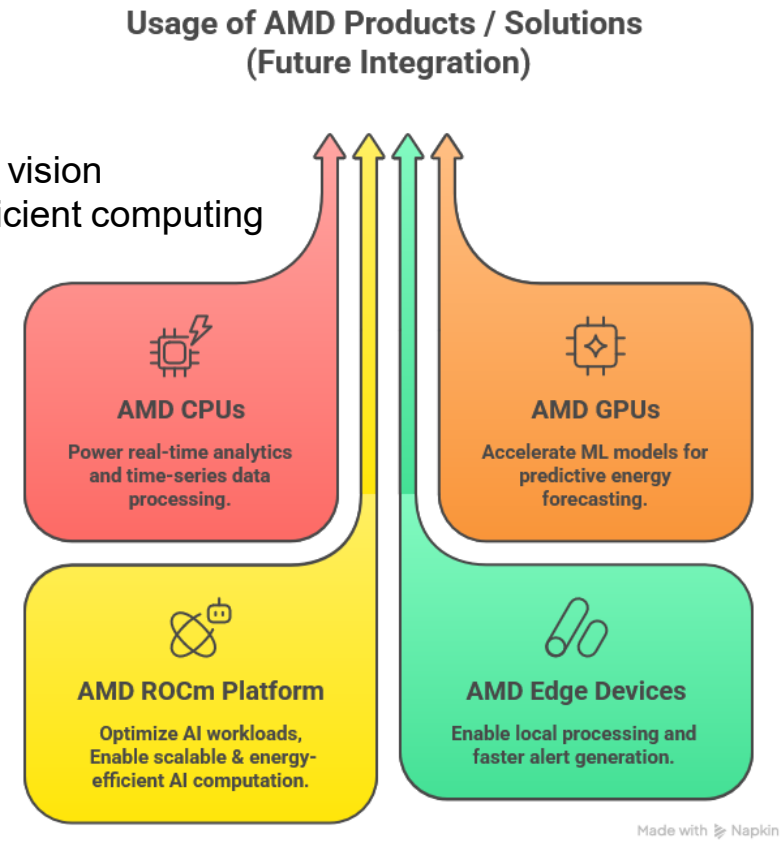


Technologies to be used in the solution



Usage of AMD Products

EcoNudge AI aligns with AMD's vision
of high-performance, energy-efficient computing



Estimated implementation cost

Implementation Cost Comparison

Cost Component	Initial Prototype	Future Full-Scale Deployment
Software Development	Low cost	Optimized cost-performance
Tools	Free	Medium
Cloud Hosting	Minimal	Medium
IoT Sensors & Smart Meters	Not applicable	Medium
AMD-Powered Compute Resources	Not applicable	Optimized cost-performance
Maintenance & Upgrades	Not applicable	Moderate

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Cost-effective with high long-term savings through reduced energy & water wastage

Prototype Assets

➔ GitHub Public Repository Link

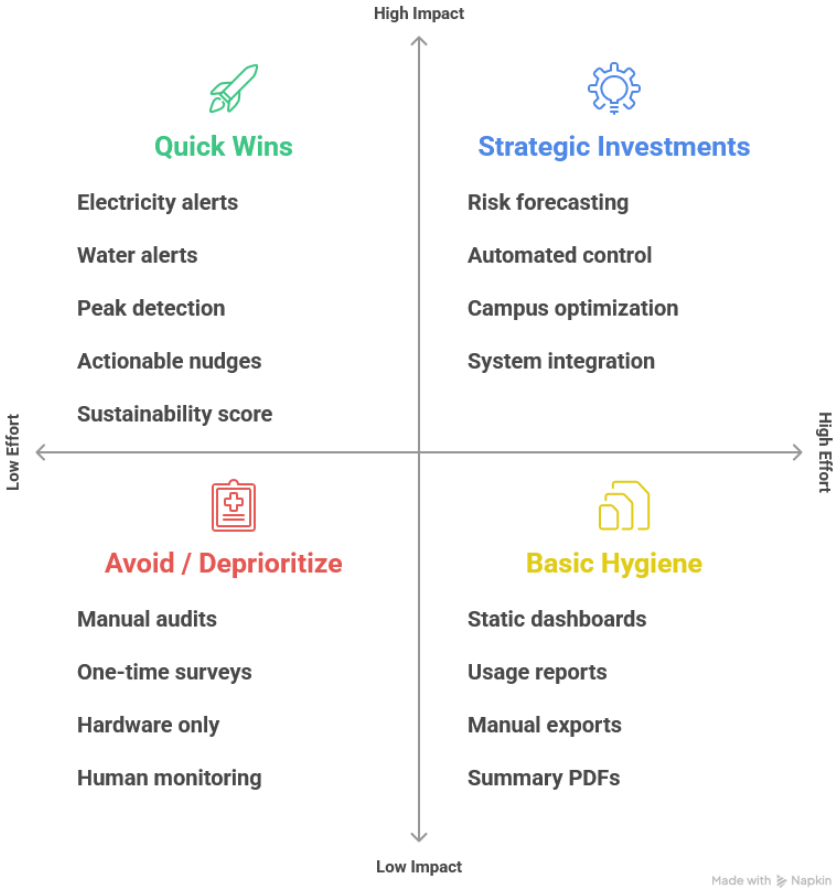
SindamMeghaSree/EcoNudge-AI: EcoNudge-AI is a smart campus sustainability dashboard built with Streamlit and Python. It tracks electricity and water usage across campus locations, generates real-time alerts, and computes a Campus Sustainability Score, CO2 Score, savings. The dashboard features dynamic graphs and charts to visualize trends and comparisons..

➔ Demo Link

https://drive.google.com/file/d/1AsWf866I4dO8qwnfRKqIOHmoups3STjL/view?usp=sharing

Impact Matrix(Effort vs Impact)

Axes
•X-Axis: Effort to Implement (Low → High)
•Y-Axis: Impact on Sustainability (Low → High)



Researches and References

Energy Management

Smart grid and energy management research papers.

Water Leakage

Water distribution and leakage detection studies.

Sustainability Frameworks

Sustainability and ESG frameworks.

Anomaly Detection

AI-based anomaly detection literature.

Energy Efficiency Reports

Government and global energy efficiency reports.

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AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

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Thank you!